HipMRI 3D Improved U Net Feasibility Review Version 5

Midpoint proposal for a patient separated 3D comparison supporting reviewable prostate contour drafting

DATASET	DIFFICULTY	BASELINE	PROPOSED MODEL	TARGET
CSIRO HipMRI **211 scans / 38 patients**	Hard	Plain 3D U Net	Residual and deep supervised 3D U Net	Patient mean prostate Dice at least 0.70

User need and engineering decision

A radiotherapy contouring specialist needs a human reviewed draft of prostate and organ at risk boundaries, not an autonomous treatment decision. The question is whether 3D context, residual refinement and deep supervision reduce small structure and cross slice errors enough to justify added memory, runtime and complexity. Earlier 2D OASIS work establishes a reusable workflow; it does not provide HipMRI performance evidence.

Scope and five acceptance criteria

1. Zero patient overlap between train, validation and test. 2. MRI and mask pairs pass pairing, shape, affine, finite value and integer label checks. 3. Held out patient mean prostate Dice is at least 0.70, with per class Dice and IoU. 4. Parameter count, peak GPU VRAM and per case inference time are recorded under matched settings. 5. Three to five held out failures are reviewed with MRI, ground truth, prediction, prostate false positives and false negatives.

HipMRI dataset and model rationale

The CSIRO collection Labelled weekly MR images of the male pelvis provides longitudinal pelvic MRI and six indexed masks. Its description states 5 = Prostate; this provenance defines the target. The seed 42 manifest assigns 27/6/5 patients and 146/46/19 scans to train/validation/test, keeping all weekly scans from one patient together. MRI intensities are z score normalised; images use trilinear resizing and masks nearest neighbour resizing. A plain 3D U Net is the course aligned baseline. The proposed model adds residual units and two auxiliary decoder heads used only during training.

Literature collected by midpoint and how it changes the plan

Source	Relevant finding	Project decision
Çiçek et al. 2016	3D convolutions and encoder decoder skips learn volumetric context.	Use a plain 3D U Net as the matched baseline.
Isensee et al. 2018	Residual refinement and deep supervision support 3D medical segmentation.	Use compact residual blocks and training only auxiliary heads; do not claim exact reproduction.
Dowling et al. 2015	Repeated male pelvis MRI is relevant to radiotherapy contouring research.	Split by patient, preserve weekly scans and treat output as review support.
CSIRO description	Defines six labels and identifies label 5 as prostate.	Validate integer labels and record class provenance.

Preliminary feasibility evidence available at midpoint

The data audit identifies 211 paired scans from 38 patients and confirms zero patient overlap. Local synthetic NIfTI tests exercise recursive pairing, invalid data rejection, patient separation, finite 3D forward and backward passes, auxiliary outputs and validation only checkpoint selection. The baseline and improved modules, loss, training loop and prediction path are implemented. Appendix A shows the reusable 2D workflow; Appendix B records the HipMRI specific adaptation and literature to design traceability.

Risks compute budget next experiment and fallback

Risks are 3D memory pressure, prostate under sampling in random crops, orientation or affine errors, slow NIfTI I/O, overfitting across 38 patients and failed cluster jobs. The budget is one short Rangpur GPU smoke run followed by two matched full runs, one plain and one improved, on one GPU at a time. Start with batch size 1; enable mixed precision after the smoke check; select checkpoints on validation data only; do not inspect held out cases during model choice. If the improved model provides no reliable boundary benefit at acceptable cost, retain the plain 3D U Net.

Questions for tutor confirmation

1. Does the matched plain versus residual and deep supervised 3D study satisfy Hard scope? 2. Is patient level prostate Dice the correct headline aggregation when patients have different weekly scan counts? 3. Is the GPU budget and 96 cube training crop strategy proportionate?

References [1] Çiçek et al. 3D U Net. MICCAI 2016. arXiv 1606.06650. [2] Isensee et al. Brain Tumor Segmentation and Radiomics Survival Prediction. 2018. arXiv 1802.10508. [3] Dowling et al. Int J Radiat Oncol Biol Phys 93 5. 2015. DOI 10.1016/j.ijrobp.2015.08.045. [4] CSIRO. Labelled weekly MR images of the male pelvis. v2. DOI 10.25919/45T8-P065.

Appendix A 2D Implementation Evidence

Supporting evidence for the reusable 2D workflow and not HipMRI performance

The 2D stage verifies the segmentation training and evaluation workflow. The HipMRI stage reuses that engineering experience while introducing a new dataset, clinical target, 3D data pipeline and 3D models. Trained 2D weights and OASIS scores are not transferred as HipMRI evidence.

Evidence retained from the 2D implementation

Dataset	Training	Best validation DSC	Test foreground mean DSC
OASIS brain MRI slices	80 epochs on A100	0.9770 at epoch 29	0.9694

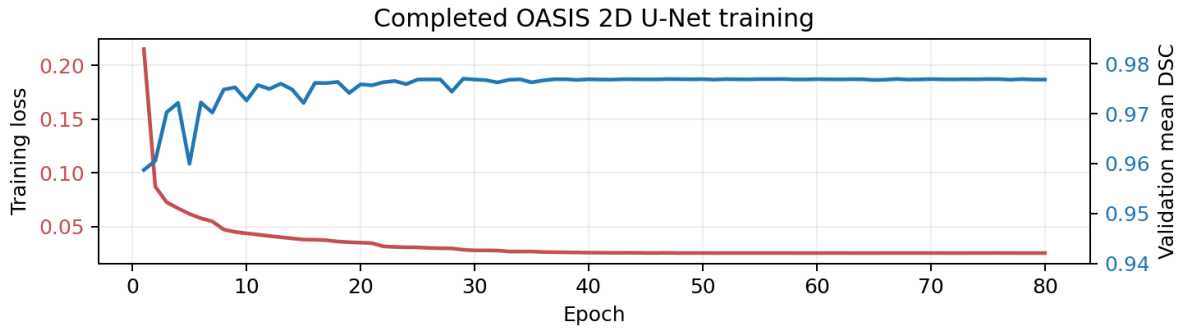


Figure 1 Training loss decreases while validation mean DSC stabilises near 0.977

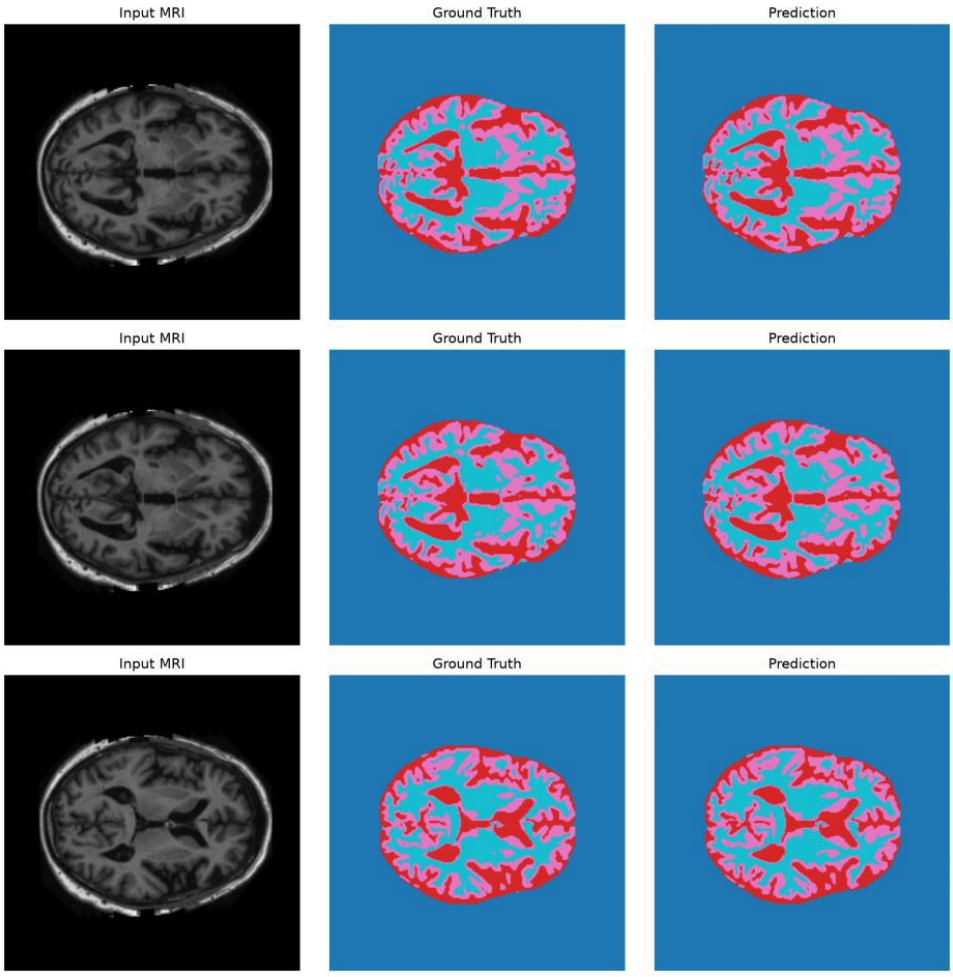


Figure 2 Held out OASIS examples showing input MRI ground truth and 2D U Net prediction

Appendix B HipMRI Literature and 3D Implementation Plan

Supporting detail for the midpoint review with HipMRI evidence kept separate from 2D workflow evidence

What was learned and how it was applied

Reading or source	Collected point	Concrete implication for HipMRI
3D U Net Çiçek et al.	Volumetric kernels and skips combine through plane context with local detail.	Use B 1 D H W input, three encoder stages, skips and six logits. Keep it as baseline.
Residual and deep supervision Isensee et al.	Residual refinement and auxiliary supervision can improve deep 3D optimisation.	Add residual shortcuts and two training only heads; final inference uses one head; compare under identical settings.
HipMRI context Dowling and CSIRO	Repeated pelvic MRI supports contouring research but creates patient leakage risk.	Group weekly scans by patient; report patient mean prostate Dice; frame output as a reviewed draft.
Small foreground	The prostate can be overwhelmed by background and larger pelvic structures.	Use prostate weighted cross entropy plus soft Dice and 0.5 prostate containing crop probability.
Evaluation design	Aggregate scores can hide local omissions and excess contours.	Pre register per class Dice and IoU, resource measures and 3 to 5 paired FP and FN panels; select on validation only.

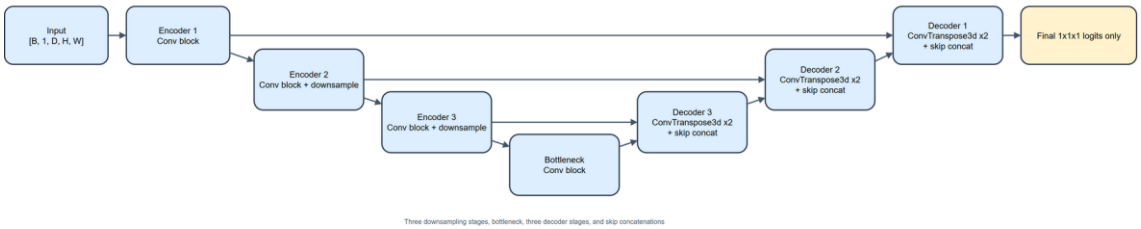
HipMRI specific implementation status at midpoint

Stage	Midpoint status	Evidence or next action
Data audit and pairing	Implemented	211 MRI and mask pairs; 38 patients; affine, shape, finite value and integer label checks.
Patient split	Frozen	Seed 42; 27/6/5 patients; 146/46/19 scans; zero overlap.
3D models	Implemented	Plain baseline and compact residual improved model with two training only auxiliary heads.
Local verification	Passed	Synthetic pairing, patient split, forward and backward, loss, checkpoint reload and prediction path.
Real data evaluation	Pending	Rangpur data audit, GPU smoke test, then two matched full experiments.

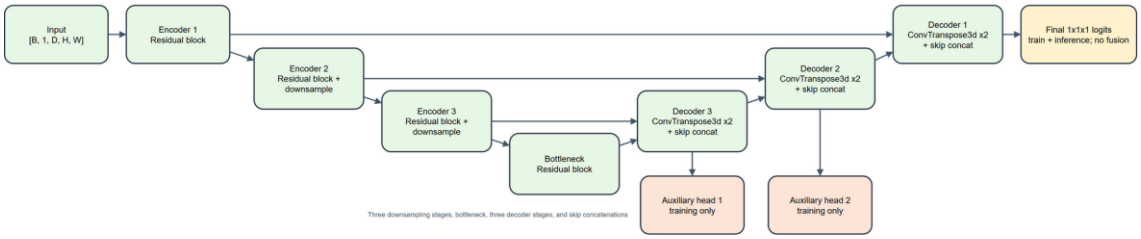
Planned matched experiment

1 Audit HipMRI	2 Freeze manifest	3 GPU smoke run	4 Train both arms	5 Evaluate once
Pairing and labels	Patient disjoint	Memory and I/O	Same protocol	Metrics and failures

Baseline 3D U-Net



Historical Improved (improved_v1; 4232e0d)



Historical decoder: ConvTranspose3d x2 -> skip concatenation -> Conv/Residual block. Auxiliary heads add training losses only; no inference logits fusion.

Matched controls

Use the same manifest, target shape, crop policy, optimiser, learning rate, epoch budget, validation only checkpoint rule and held out evaluation. Record parameters, training and inference VRAM, and inference time.

Fallback

Retain the plain 3D U Net if the proposed model does not deliver a reviewable boundary benefit at acceptable cost. Do not use OASIS scores, synthetic test success or test set inspection to choose the HipMRI model.

Evidence boundary Appendix A demonstrates a reusable 2D engineering workflow. Appendix B documents the HipMRI specific design and planned evidence. Neither appendix claims completed HipMRI performance at midpoint.